

José Hiram Soltren

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SYNOPSIS

Staff Software/Hardware Engineer: from Silicon through Display to Web.
Seeking Hardware Integration, Firmware, Scientific Computing, Infra roles.
Flight Instructor - CFI/CFII/AIGI (MEI pending)
Expert in C, C++, systems, Linux, device drivers, hardware accelerators.

EXPERIENCE

Flight Instructor Soltren Consulting LLC Apr 2024 - present

- First class medical. 626h total, 518h PIC, 274h XC, 51h IMC, 125 dual given.
- Commercial pilot, single/multi engine airplane.

Connectivity Engineer ForeFlight Jan 2024 - Nov 2024

- Designed and implemented a high-performance UDP-based protocol and API for generic third-party avionics integration, successfully adopted in production by external partners (e.g., uAvionix).
- Engineered a highly optimized payload structure utilizing a novel 4-bit integer format to encode ARINC 424 flight plans, maximizing data transfer efficiency and reliability.
- Resolved a critical, FAA-escalated weather parsing outage by rapidly debugging and fixing a decade-old database query defect within one week of taking ownership, restoring safe flight operations for offshore helicopter fleets.
- Spearheaded the technical infrastructure for an on-site avionics lab, physically integrating hardware from multiple manufacturers to rapidly diagnose and deploy fixes for complex flight plan and data transfer bugs.

Team Lead/Manager, Build and Release Roku Jan 2023 - Sep 2023

- Responsible for official build, release, and CI infrastructure.
- Led team of four to implement self-service efforts, reducing ticket counts by 30% while improving turnaround time and throughput.
- Implemented pre-merge CI, presented results at IEEE ICST 2023.

Senior Software Engineer, Roku Players Roku Jun 2020 - Dec 2022

- C++20 based firmware development with focus on video decode and multimedia.
- YouTube certification lead for HDMI player platforms.
- Firmware lead for Roku Streaming Stick 4K, a 20M unit program.

Deep Learning Software Engineer Nervana / Intel Corp. Aug 2017 - Feb 2019

- Firmware development in C++ for deep learning training hardware accelerator.
- Focus on hardware and software to run GEMM and convolution as fast as possible.
- Designed an algorithm for all-to-all data transfers on our hardware that beat the Bruck Algorithm, the fastest general purpose algorithm.

Software Engineer Cloudera, Inc. May 2016 - Aug 2017

- Fault tolerance hardening for Apache Spark. Presented findings at Spark Summit.

Senior System Software Engineer NVIDIA Corporation May 2011 - Apr 2016

- Firmware and driver development for 3D graphics accelerators.
- Lead engineer for video decode and post-processing, including hardware validation.
- Bug fix and feature work in OpenGL and Vulkan.

EDUCATION	Massachusetts Institute of Technology, Cambridge, MA <i>Master of Engineering and S.B., Electrical Engineering and Computer Science</i> Grad level operating systems course work. Build a kernel and hypervisor in C. Lab assistant for undergraduate microcontroller/firmware course.
CITIZENSHIP	United States